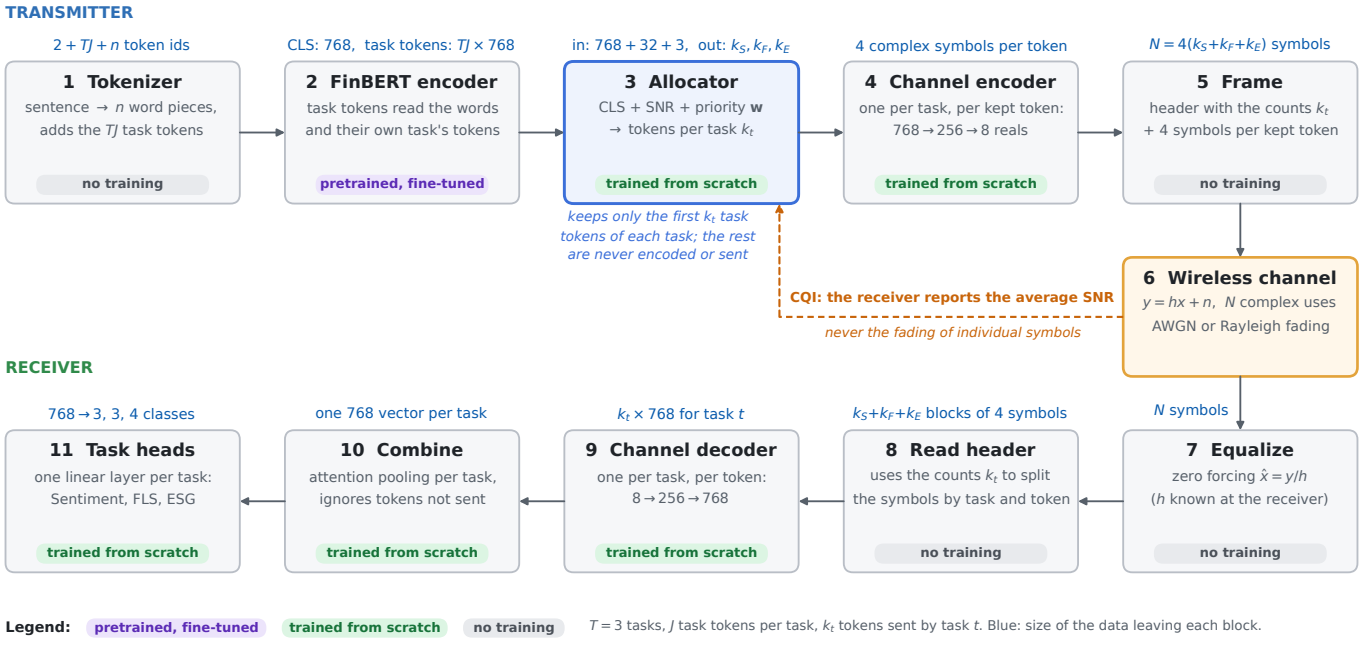
TRANSMITTER
2 + TJ + n token ids
1 Tokenizer
sentence → n word pieces, adds the TJ task tokens
no training
CLS: 768, task tokens: TJ × 768
2 FinBERT encoder
task tokens read the words and their own task's tokens
pretrained, fine-tuned
in: 768 + 32 + 3, out: k_S, k_F, k_E
3 Allocator
CLS + SNR + priority w → tokens per task k_t
trained from scratch
keeps only the first k_t task tokens of each task; the rest are never encoded or sent
4 complex symbols per token
4 Channel encoder
one per task, per kept token: 768 → 256 → 8 reals
trained from scratch
$N = 4(k_S+k_F+k_E)$ symbols
5 Frame
header with the counts k_t + 4 symbols per kept token
no training
6 Wireless channel
$y = hx + n$, N complex uses AWGN or Rayleigh fading
CQI: the receiver reports the average SNR
never the fading of individual symbols
RECEIVER
N symbols
7 Equalize
zero forcing $\hat{x} = y/h$ (h known at the receiver)
no training
$k_S+k_F+k_E$ blocks of 4 symbols
8 Read header
uses the counts k_t to split the symbols by task and token
no training
$k_t \times 768$ for task t
9 Channel decoder
one per task, per token: 8 → 256 → 768
trained from scratch
one 768 vector per task
10 Combine
attention pooling per task, ignores tokens not sent
trained from scratch
768 → 3, 3, 4 classes
11 Task heads
one linear layer per task: Sentiment, FLS, ESG
trained from scratch
Legend: pretrained, fine-tuned trained from scratch no training
T = 3 tasks, J task tokens per task, k_t tokens sent by task t. Blue: size of the data leaving each block.